Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_24

1. Summary of Key Deviations

The analysis of Run_24 reveals a major physical fault characterized by a severe downstream blockage, leading to a massive redistribution of fuel flow.

- **Flow_Bypass (Major):** The most significant anomaly is a massive positive **offset**, with bypass flow reading up to **+59.7%** higher than healthy. This indicates the vast majority of the fuel is being diverted away from the engines.
- **Flow_Engine1-4 (Moderate):** All four engine flows show a major, consistent **negative offset** of approximately **-25%**. This confirms that fuel is being severely restricted from reaching the combustion chamber.
- **System-Wide Pressure Loss (Moderate):** Most system pressures (`p_Combustion`, `p_FMU`, `p_Shut`) show a moderate **negative offset**, dropping by **-10% to -30%**.
- **Flow_Pump & Flow_PRV (Nominal):** The pump flow and PRV flow show only minor or negligible deviations, indicating the pump itself is healthy and the system is not reaching a critical over-pressurization state.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Fuel Metering Unit (FMU)**	**Severe Partial Blockage / Restricted Opening**	This is the most likely root cause. The combination of **very low engine flow** (`Flow_Engine` is -25%), **extremely high bypass flow** (`Flow_Bypass` is +60%), and **nominal pump flow** creates a classic signature of a major downstream restriction. The FMU is the primary component capable of causing such a uniform blockage across all four engine lines, forcing the bypass loop to handle the diverted flow.

2	**Injector Nozzles (All Four)**	**Simultaneous Partial Clogging**	It's possible that all four injector nozzles became partially clogged at the same time. This would produce the same symptoms as a blocked FMU. However, a single point of failure in the upstream FMU is generally more probable than four simultaneous, identical failures in the nozzles.
3	**Shut-off Valve**	**Partial Blockage**	A partial blockage in the shut-off valve, which is downstream of the FMU, could also cause a system-wide restriction. It is another plausible location for the blockage.
4	**Bypass Valve**	**Incorrect Calibration / Sticking**	The bypass valve is opening very wide to divert the flow that cannot pass the blockage. This is a *symptom* of the blockage, not the root cause. The valve is reacting correctly to the high back-pressure.
5	**System Contamination**	**Debris causing blockage**	This is the likely underlying *cause* of the blockage itself (e.g., a piece of debris lodging in the FMU). The primary diagnostic task is to identify the *location* of the blockage, which is clearly downstream of the bypass tee and before the engine nozzles.

3. Most Probable Hypothesis

The most probable root cause is a **Severe Partial Blockage within the Fuel Metering Unit (FMU).**

Reasoning: The data provides a clear, textbook case of a major downstream restriction. The pump is operating nominally, but the fuel is not reaching the engines. The three key symptoms—**severely decreased engine flow (-25%),**

massively increased bypass flow (+60%), and nominal pump flow—all point to a single conclusion: something is preventing the majority of the fuel from getting past the point where the bypass loop tees off. The FMU is the most logical single point of failure that would affect all four engine nozzles simultaneously and uniformly. The system is correctly detecting the resulting high back-pressure and diverting the pump's output through the bypass loop.